

POSITION, NOT CONTENT: WHAT REPAIRS QUESTION-FIRST PROMPTING IN VISION-LANGUAGE MODELS

Anonymous authors

Paper under double-blind review

ABSTRACT

Putting the question before the image rather than after it changes nothing about what a vision-language model is told, yet it changes what the model answers. Prior work by the present authors established this question-first paradox on three yes/no benchmarks and traced it to read-out rather than perception: the question does reach the image and visibly steers it, but the answer position cannot read it back, and repeating either the question or the image repairs the damage. What that work could not say is how far the effect reaches, what the repeated image actually contributes, and what it costs. We answer those three questions. Across 14 further benchmarks the paradox proves general, costing 7 points on average and as much as 18 on a single benchmark, yet it is not a law: on object detection, where the model returns boxes instead of a word, the effect reverses and asking first is the better choice. The repeated image turns out to carry almost no information. Shrinking it to $\frac{1}{8}$ of its resolution leaves accuracy statistically unchanged while cutting its price from thousands of extra tokens to about 100, and a reversed copy works as well as a correct one, so what matters is where the tokens sit and not what they show. Breadth also marks a limit: on video, repeating the image stops helping, while repeating the question still does. Finally, a reflective prompt optimizer given the same budget improves only the ordering our diagnosis says has room left, and never catches the simpler fix, so this is not something better wording would have found.

1 INTRODUCTION

Ask a vision-language model a question about an image and you must decide, often without thinking about it, whether the question goes before the image or after it. The information content is identical either way. Under causal attention, putting the question first is the option that lets every image token see it, which is what a reader would want when told to look for something specific. It makes the model worse, and on one benchmark category it more than halves accuracy.

Relation to prior work. Prior work by the present authors (Anonymous, 2026), included in the supplementary material, established this question-first paradox on three yes/no benchmarks and diagnosed it: the question does reach the image and measurably steers its patches, but the answer position cannot read the question back out, and an attention knockout shows the two orderings answer through different paths. That work also proposed echoing as the repair. We recap the diagnosis in Section 3 because the new results are not interpretable without it, and we cite rather than re-derive it.

Three questions that work could not answer motivate this paper, and each turns out to have a sharp answer.

Does the paradox survive outside yes/no visual question answering? We run the full ordering ladder on 14 benchmarks not previously tested, including four video QA datasets and a 24,449-item detection-style presence benchmark (RF20-VL). Question-first hurts on 12 of 13 image and video benchmarks, costing 7.0 points on average, and on RF20’s yes/no presence task it hurts on 7 of 7 categories, taking Lab Imaging from 86.3 to 38.9. The same breadth exposes two boundaries

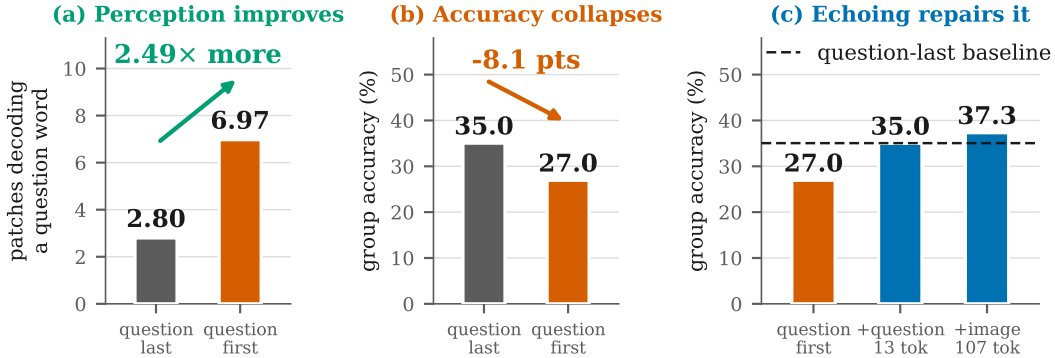


Figure 1: The question-first paradox and its repair, Qwen3-VL-8B. **(a)** Question-first makes 2.49 $\times$ more image patches decode a question content word (160 NaturalBench candidates), so the question reaches the image. **(b)** It still loses 8.1 points of group accuracy ($p = 1.3 \times 10^{-12}$). **(c)** Echoing the question after the image repairs it for 13 tokens; an $\frac{1}{8}$ -resolution image echo beats the baseline for 107.

the original three benchmarks could not. On video, where the echo doubles an already long frame sequence, image echoing loses to question-last on three of four datasets. On RF20-VL detection, where the model emits boxes rather than an answer token, the paradox itself reverses: question-first is 2.80 mAP *better* and wins on 17 of 20 datasets, which places the effect in read-out rather than in prompt order as such.

Which patches change, and can we find them without cherry-picking? The qualitative evidence for steering has so far been chosen by hand, one persuasive image at a time, which is exactly the kind of evidence a reader should distrust. We replace that with a procedure. For a given example we decode every image patch under both orderings with a logit lens, drop the patches whose top-1 token is unchanged, and rank what survives by whether the new token belongs to the question’s own vocabulary. Nothing is selected by eye: the figure reports every patch that changed, and the ranking, not the author, decides which ones are worth naming. Figure 1 is that output.

What is the echo actually made of? If the echo re-anchors read-out rather than supplying a second look, it should survive being destroyed as an image. It does. Downscaling the echoed copy to $\frac{1}{8}$ resolution, $\frac{1}{64}$ of the pixels, leaves NaturalBench group accuracy statistically indistinguishable from a full-resolution echo (37.32 vs 37.37, $p = 1.00$) while still beating question-last ($p = 0.005$), and it cuts the price from 3,352 extra tokens per query to 107. RF20 is the lone exception, and an informative one: its targets are small, so there the second look is doing real perceptual work and degrading it costs 1.76 F1.

Would careful prompt engineering do the same job? We give GEPA (Agrawal et al., 2025), a reflective prompt optimizer, the same budget under both orderings. It improves question-first on all three benchmarks it is run on and finds nothing at all under question-last, which is what the read-out diagnosis predicts, since only the broken ordering has headroom to recover. It closes at most 83% of the gap and never reaches the plain question-last baseline.

Together these say something the original result did not: the repair is not about giving the model a better look at the image, it is about where the tokens sit, it costs about a hundred tokens rather than a doubled prompt, and it cannot be replaced by rewording. We also report where it fails, which is video and any task whose evidence is small enough that a degraded second look loses it.

2 SETUP

Notation. A prompt has three sections: the system message (S), the image (I), which expands to many visual tokens, and the task or question (T). An ordering is written as its sequence, so SIT is system-image-question and STI is system-question-image. An ordering may repeat a section: SITIT repeats the question after the image, SITIT repeats the image after the question, and SITIT_{rev} is SITIT with the second image copy spatially reversed. The two baselines are SIT, which is what

most inference code emits by default, and *STI*, which is what a reader is told to do when looking for something specific.

Models. Qwen3-VL-8B is the primary model and carries every analysis. For breadth we also run Gemma-3-27B, Gemma-4-31B, Qwen2.5-VL-7B, InternVL3-8B and LLaVA-1.5, with the Gemma models loaded in 4-bit on a single GPU. All results use greedy decoding with a fixed answer-token budget, and every ordering of a given benchmark shares one system message, one question string per item, one image preprocessing path and one scoring function, so the ordering is the only difference between columns.

Benchmarks. Three benchmarks carry the diagnosis recapped in Section 3: NaturalBench (Li et al., 2024a), 1,900 groups (7,600 pairs); POPE (Li et al., 2023), 9,000 items over its random, popular and adversarial splits; and Winoground (Thrush et al., 2022), 400 examples (1,600 items). The generality study of Section 4 adds fourteen benchmarks not tested in prior work: RF20-VL, used in two forms, as an object-detection benchmark scored by mAP over 20 datasets and 3,970 images with multi-class prompting, and as a 24,449-question yes/no presence task over the same images in 7 categories; nine image benchmarks (BLINK (Fu et al., 2024), CV-Bench (Tong et al., 2024), HR-Bench, MMStar (Chen et al., 2024), MMVP, RealWorldQA, WorldMedQA-V, TallyQA (Acharya et al., 2019), VQAv2 (Goyal et al., 2017)); and four video QA benchmarks (MVBench (Li et al., 2024b), NExT-QA (Xiao et al., 2021), MSVD-QA (Xu et al., 2017), TGIF-QA (Jang et al., 2017)). For NaturalBench and Winoground we pair on the group or example index and report an exact two-sided McNemar test alongside a paired bootstrap 95% confidence interval over 5,000 resamples.

Interpretability tools. All mechanistic analyses use a logit lens (nostalgebraist, 2020; Belrose et al., 2023), projecting intermediate residual-stream states through the unembedding to read out per-layer top tokens, either at the answer position or over image patches. The attention knockout follows the edge-severing methodology used for factual recall in language models (Geva et al., 2023; Meng et al., 2022).

3 BACKGROUND: THE PARADOX AND ITS DIAGNOSIS

This section recaps the result this paper extends (Anonymous, 2026). All numbers here are reproductions on our own runs; the claims are prior work.

3.1 QUESTION-FIRST PERCEIVES BETTER AND ANSWERS WORSE

Table 1 runs six orderings on four benchmarks with Qwen3-VL-8B. Every column uses the same system message, question string, image, decoding and scoring code; *STI* and *SIT* contain the same tokens in the same quantity. Only the position of the question moves. Question-first is the worst ordering on all four: NaturalBench group accuracy falls from 35.05 to 26.95 ($p = 1.3 \times 10^{-12}$), Winoground from 31.75 to 22.25 ($p = 7.7 \times 10^{-5}$), RF20 F1 from 78.57 to 65.59.

The obvious explanation, that question-first degrades perception, is wrong. Reading image patches with a logit lens, question-first makes 6.975 patches per image decode a question content word against 2.800 for question-last, a factor of 2.49 over 160 NaturalBench candidates. Figure 2 shows this patch by patch: asking “Is the adult in the image chasing a child?” before the image turns *running* into *chase*, *boy* into *ahead* and *baseball* into *child*, taking the count of patches decoding a question word from 1 to 12, and the model then answers “No” to a question whose ground truth is Yes.

3.2 LOCATING THE REWRITE AUTOMATICALLY

The patch-level evidence above is produced by a procedure rather than by inspection, which matters because the alternative is choosing the persuasive image and hoping it is typical. Given an example and a layer, we run the model under both orderings, project every image patch through the unembedding, and take the top-1 token per patch. Patches whose token is unchanged are dropped. Of those that remain, we mark the ones whose new token belongs to the question’s vocabulary, matched by

Table 1: Ordering ladder on four benchmarks, Qwen3-VL-8B, reproducing Anonymous (2026) and adding RF20. STI (question first) is worst everywhere. POPE is the exception to the repair: plain question-last already wins there. *N*: NaturalBench 1,900 groups, POPE 9,000, Winoground 400 examples, RF20 24,449.

Benchmark	Metric	IST	SIT	STI	STIT	SITIT	SITIT_rev
NaturalBench	pair acc.	78.93	79.21	75.29	78.95	80.36	80.26
NaturalBench	group acc.	33.95	35.05	26.95	35.00	37.37	37.37
POPE	F1	88.22	88.35	86.13	87.61	88.12	88.25
Winoground	group acc.	28.00	31.75	22.25	37.50	40.25	41.00
RF20	F1	78.92	78.57	65.59	78.61	80.56	79.78

Q: “Is the adult in the image chasing a child?” (ground truth: Yes)

ringed cells decode to a word from the question (adult, chasing, child); Qwen3-VL-8B, layer 32

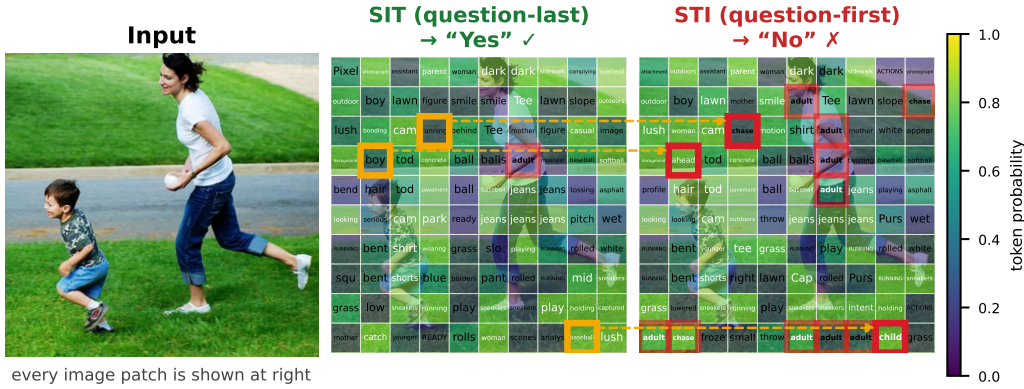


Figure 2: Steering happens, and is not read out. Every image patch is shown: each cell holds the vocabulary token that patch’s layer-32 hidden state decodes to under the logit lens, shaded by that token’s probability. Red rings mark patches decoding a word from the question. Question-last leaves the patches generic (1 ringed patch) and answers correctly; question-first steers the same patches onto the question’s vocabulary (12 ringed patches) and answers wrong. Amber boxes mark three individual patches in the question-last panel, joined by a dashed arrow to the same patch in the question-first panel, where it is boxed in red: *running* becomes *chase*, *boy* becomes *ahead*, and *baseball* becomes *child*. Echoing the question keeps all 12 steered patches and answers correctly again.

stem against the question’s content words and their near variants, so that a patch turning *throws* into *chase* counts and one turning *lush* into *grass* does not. The result is a count with a denominator, reported in every figure caption, rather than a gallery.

Two properties make this usable as evidence. It is exhaustive, so the figure shows every patch that changed and a reader can check the ones we did not name. And it is cheap, needing two forward passes and no training, so it runs on any example rather than the handful that happen to look good. We ship it as an interactive viewer in the supplementary material, which steps through the changed patches one at a time; the paper figures are single frames of it.

3.3 THE FAILURE IS READ-OUT, AND IT IS CAUSAL

Probing the answer position on the 150 NaturalBench pairs where question-first is wrong and question-last is right, question-first sends $2.19\times$ less attention to the question at the peak read-out layer (0.0676 vs 0.1479) and over twice as much to the image. The two orderings are indistinguishable at chance through layer 25 of 36, then fork at layer 26 and reach $P(\text{correct})$ of 0.0986 and 0.8747 (Figure 3).

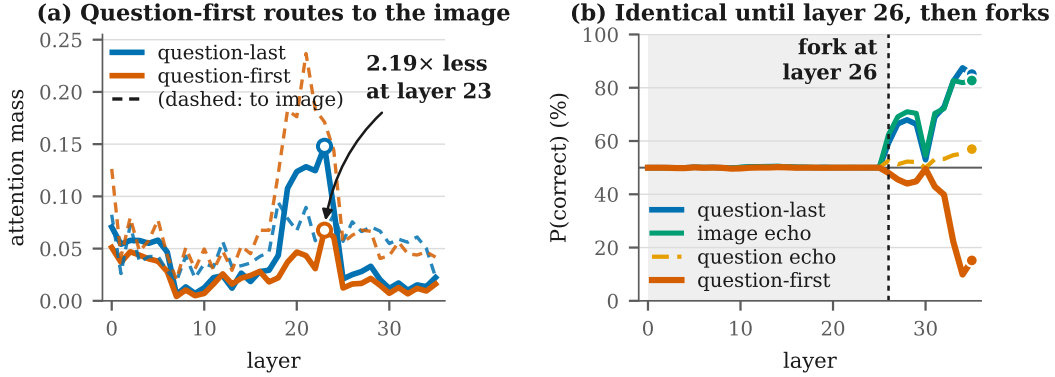


Figure 3: Where question-first breaks (150 NaturalBench pairs where it is wrong and question-last is right). **(a)** Attention leaving the answer position, to the question (solid) and the image (dashed): at layer 23 question-first sends $2.19\times$ less to the question. **(b)** $P(\text{correct})$ per layer. The two baselines sit at chance through layer 25, fork at 26 and never re-cross; both echo orderings track question-last.

An attention knockout makes the dependence causal and doubly dissociated (Appendix B): severing answer-to-question costs question-last 5.6 points (95% CI $[+1.99, +9.60]$) and question-first nothing (+0.4), while severing answer-to-image costs question-first 9.6 points and question-last nothing. An equal-length random span moves both by at most 0.8.

3.4 THE REPAIR, AND WHAT IT LEAVES OPEN

The diagnosis prescribes the prompt: if the answer position cannot read the question, put something after the image that it can read. Question echoing (STIT) repeats the question after the image, which restores question-answer adjacency while provably leaving perception untouched, since causal masking means a later copy cannot alter earlier patches ($\cos(\text{STI}, \text{STIT}) = 0.99999$). Image echoing (SITIT) repeats the image after the question. Prior work also showed that reversing the echoed copy (SITIT_{rev}) changes little, which we reproduce: NaturalBench group accuracy moves from 37.37 to 37.37 ($p = 1.00$) and Winoground from 40.25 to 41.00 ($p = 0.68$).

That work listed the second image’s token cost as an open limitation and did not measure it. Sections 4 to 5.3 take up that thread: what the echo costs, how little of it is needed, where the repair stops working, and whether a prompt optimizer would find it anyway.

4 GENERALITY AND BOUNDARIES

Prior work tested three yes/no benchmarks. That leaves open whether the paradox is a property of yes/no visual question answering, of those three datasets, or of the prompt layout itself. We run the full ordering ladder on 13 further benchmarks spanning single-image perception, high-resolution perception, counting, medical VQA and four video QA datasets, and separately on RF20, a 24,449-item presence-detection benchmark built from 20 detection datasets across 7 categories.

4.1 THE PARADOX HOLDS ON 13 NEW BENCHMARKS

Moving the question from after the image to before it costs Qwen3-VL-8B 7.01 points on average and hurts on 12 of 13 (Table 2). The effect is far larger than the differences that usually separate published models: CV-Bench loses 18.01 points, MMStar 11.40, MMVP 9.00, NExT-QA 8.91. WorldMedQA-V is the single benchmark where question-first is not worse (+1.06); we report it rather than dropping it, and note it is also the smallest benchmark in the set ($N = 568$).

The two echo orderings separate the cost of the repair from its size. Question echoing recovers 5.49 of the 7.01 points, or 78% of the deficit, for 13 extra tokens. Image echoing recovers all of it and exceeds question-last by 1.18 points, winning on 9 of 13.

Table 2: Full ordering ladder for Qwen3-VL-8B across 13 benchmarks not tested in prior work, video QA below the rule. Best of the four orderings in bold. The echo column gives SITIT with the second image copy at half resolution. Question-first hurts on 12 of 13; image echoing recovers the loss on the image benchmarks and does not on video.

Benchmark	N	Ordering				Echo $\frac{1}{2}$ SITIT	Δ vs SIT	
		SIT	STI	STIT	SITIT		STI	SITIT
BLINK	793	53.09	44.51	52.21	53.97	50.44	-8.58	+0.88
CV-Bench	2,638	81.35	63.34	79.98	83.55	83.70	-18.01	+2.20
HR-Bench	1,600	67.81	59.25	63.94	69.75	68.44	-8.56	+1.94
MMStar	1,500	53.33	41.93	51.47	55.13	56.07	-11.40	+1.80
MMVP	300	73.00	64.00	72.33	77.67	76.00	-9.00	+4.67
RealWorldQA	765	69.28	63.92	67.19	70.33	70.33	-5.36	+1.05
WorldMedQA-V	568	53.35	54.40	56.87	57.22	57.57	+1.06	+3.87
TallyQA	38,589	79.14	74.15	77.85	79.99	79.36	-5.00	+0.84
VQAv2	214,354	81.66	78.72	80.49	81.30	81.38	-2.94	-0.36
<i>Mean, 9 image datasets</i>							-7.53	+1.88
MVBench	2,600	64.08	55.88	62.23	64.62	66.08	-8.19	+0.54
NExT-QA	8,564	78.21	69.30	76.05	77.95	78.11	-8.91	-0.26
MSVD-QA	13,157	40.05	35.52	36.61	39.27	39.51	-4.53	-0.78
TGIF-QA	25,751	36.32	34.59	33.62	35.26	35.50	-1.74	-1.06
<i>Mean, 4 video datasets</i>							-5.84	-0.39

Finding. The paradox is a property of the layout, not of yes/no visual question answering. Across 13 benchmarks never previously tested, question-first costs 7.01 points on average and hurts on 12 of 13, with a worst case of 18.01 points on CV-Bench, while prompt content is held identical.

4.2 RF20-VL: THE SIGN FLIPS WHEN THE OUTPUT IS BOXES

Every benchmark so far asks for a short answer that the model reads out at a single position. RF20-VL asks for something else: given all class names for an image, emit bounding boxes, scored by mAP over 20 detection datasets and 3,970 images. It is the one task here whose output is localization rather than a read-out, which makes it the sharpest test of how far the paradox extends.

It does not extend. The sign reverses (Table 3). For Qwen3-VL-8B, question-first is 2.80 mAP *better* than question-last (7.94 vs 5.14), and better on 17 of 20 datasets. Gemma-4-31B agrees in the same direction at a smaller size (+0.69 mAP, better on 13 of 20). We take this as a scope limit on the paradox rather than a contradiction: when the answer is a set of boxes rather than a token the decoder must read out, stranding the question before the image costs nothing, and the extra conditioning it gives every image patch appears to help.

Echoing survives the reversal. Question echoing is the best ordering for both models (+3.34 mAP for Qwen3-VL-8B, +0.91 for Gemma-4-31B), so the recommendation we make from the yes/no benchmarks does not have to be withdrawn here even though its justification does.

One control separates this task from the others. On the yes/no benchmarks, reversing the echoed image copy was free (NaturalBench $p = 1.00$). On detection it costs 1.89 mAP and helps on only 1 of 20 datasets. Together with the resolution sweep, where RF20 was the one benchmark that degraded monotonically as the echo shrank (Section 5.1), this says the echo’s content only matters when the task needs evidence localized in the image.

We exclude Gemma-3-27B from this table: it scores 0.06 mAP under every ordering, emitting many boxes that almost never land, so ordering differences there are not interpretable. The yes/no presence variant of RF20, 24,449 questions over the same images, is reported in Appendix A; that variant behaves like the other yes/no benchmarks and question-first hurts it on 7 of 7 categories.

Table 3: RF20-VL detection, mean over 20 datasets and 3,970 images, multi-class prompting. Unlike every yes/no benchmark in this paper, question-first *beats* question-last here. Question echoing is still the best ordering; the reversed echo, which is free on yes/no tasks, is the worst.

Model	Metric	SIT	STI	STIT	SITIT	SITIT_rev
Qwen3-VL-8B	mAP	5.14	7.94	8.48	6.47	3.25
	mAP ₅₀	11.15	14.97	16.01	12.07	7.67
Gemma-4-31B	mAP	13.52	14.21	14.43	14.36	n/a
	mAP ₅₀	29.75	30.12	30.48	30.58	n/a
<i>Qwen3-VL-8B Δ vs SIT (mAP)</i>			+2.80	+3.34	+1.33	-1.89
<i>datasets improved</i>			17/20	16/20	16/20	1/20

Finding. The paradox is a property of read-out, not of prompt order as such. On RF20-VL detection, where the output is boxes rather than an answer token, question-first is 2.80 mAP *better* than question-last and wins on 17 of 20 datasets. Echoing the question remains the best ordering (+3.34 mAP), but the reversed echo, free on yes/no tasks, now costs 1.89 mAP.

4.3 WHERE THE REPAIR STOPS: VIDEO

The four video QA datasets draw a boundary the original benchmarks could not. The paradox itself is if anything more reliable there: question-first hurts on 4 of 4, by 5.84 points on average. The repair, however, splits by which thing is echoed. Question echoing still recovers 3.31 of those points. Image echoing does not: it beats question-last on only 1 of 4 video datasets and loses 0.39 points on average, against +1.88 on the nine image benchmarks.

The reason is mechanical rather than mysterious. For video the “image” is a frame sequence, so an image echo doubles a visual context that is already long, and the read-out surface it adds is buried behind far more tokens than in the single-image case. The practical rule is to echo the question on video and the image on stills.

Finding. The paradox and its repair have different domains. On video, question-first still hurts on 4 of 4 datasets (−5.84), but image echoing helps on only 1 of 4 (−0.39 on average, against +1.88 on nine image benchmarks), while question echoing still recovers +3.31. Echo the question on video, the image on stills.

5 ABLATIONS: WHAT THE ECHO IS ACTUALLY MADE OF

The diagnosis in Section 3 makes a strong claim about what the second image copy is for. It is not there to be re-perceived. It is there to put image tokens after the question so the read-out step has something local to attend to. That claim has a sharp consequence: the echoed copy should not need to be legible. We test it by degrading the echo, by pricing it, and by asking whether a prompt optimizer that can only edit text finds the same fix.

5.1 THE ECHOED COPY DOES NOT NEED TO BE LEGIBLE

We downscale only the second image occurrence in SITIT, leaving the first at full resolution, and sweep the echo scale over 1, $\frac{1}{2}$, $\frac{1}{4}$ and $\frac{1}{8}$ (Table 4). On three of four benchmarks a quarter-resolution echo matches or beats a full-resolution one: POPE moves by −0.05 F1, Winoground by +0.75 group accuracy, and NaturalBench by +0.21 pair accuracy.

Pushing to an eighth makes the point sharper. On NaturalBench, an echo carrying $\frac{1}{64}$ of the original pixels scores 37.32 group accuracy against the full echo’s 37.37, a difference of 0.001 with $p = 1.00$, while still beating question-last by +0.023 ($p = 0.005$). The echo can be destroyed as an image and keep its entire benefit.

RF20 is the exception, and it is the informative one. Its accuracy falls monotonically as the echo degrades (−0.50 F1 at half, −1.76 at quarter). RF20 asks whether a specific, often small object class is present in a detection-style image, so the second look is doing real perceptual work there rather

Table 4: Echo-resolution sweep, Qwen3-VL-8B. Only the *second* image occurrence is downscaled; the first stays at full resolution. Three of four benchmarks are flat or better at quarter resolution. RF20, whose targets are small, degrades monotonically. Metrics are F1 for POPE and the RF20 presence task, group accuracy for Winoground, pair accuracy for NaturalBench. Blank cells were not run.

Benchmark	Echo 1	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$
POPE	88.12	88.35	88.07	
Winoground	40.25	41.75	41.00	41.75
NaturalBench	80.36	80.55	80.57	80.18
RF20 (presence)	80.56	80.06	78.80	

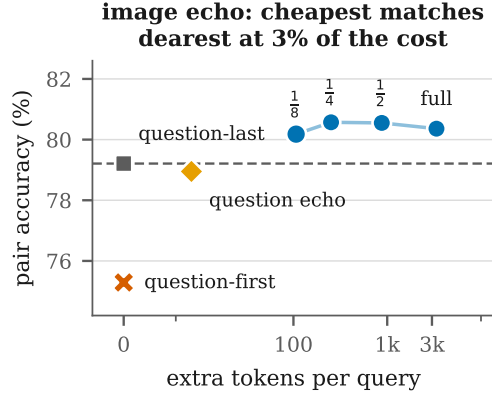


Figure 4: Cost against accuracy on NaturalBench. The x axis is extra tokens per query versus question-first (symmetric log, so 0 is shown); the dashed line is question-last. Every image echo clears that baseline, and the $\frac{1}{8}$ -resolution echo (+107 tokens) is indistinguishable from the full one (+3,352).

than only re-anchoring. The split between RF20 and the other three says the echo has two separable jobs, and that only one of them needs pixels.

Finding. The echoed image does not need to be legible. An eighth-resolution echo, carrying $\frac{1}{64}$ of the pixels, is statistically indistinguishable from a full-resolution one on NaturalBench (37.32 vs 37.37 group accuracy, $p = 1.00$) while still beating question-last ($p = 0.005$). Only RF20, whose targets are small, needs the pixels (-1.76 F1 at quarter resolution).

5.2 WHAT ECHOING COSTS

Prompt-level interventions are usually reported without their bill. We count exact token counts for each ordering with the Qwen3-VL-8B processor over 20 images spanning RF20’s constituent datasets, separating vision from text tokens (Table 12 in Appendix G).

The two echo mechanisms are three orders of magnitude apart. Repeating the question costs 13 tokens. Repeating the image at full resolution costs 3,352, which roughly doubles the prompt. Downscaling the echo collapses that cost quickly, because vision-token count falls with area: a half-resolution echo costs 875 added tokens, a quarter-resolution echo 252, and an eighth-resolution echo 107. Figure 4 puts the two axes together. The accuracy curve is flat across a $31\times$ range of echo cost, so the operating point to ship is the cheapest one.

Finding. Echo cost is a tunable knob, not a fixed doubling of the prompt. A full-resolution image echo adds 3,352 tokens per query, but a quarter-resolution echo adds 252 (7.5%) and an eighth-resolution echo adds 107 (3.2%) for indistinguishable accuracy, while question echoing adds 13.

5.3 PROMPT OPTIMIZATION FINDS PART OF THE FIX, AND ONLY WHERE THE MECHANISM SAYS IT SHOULD

A reviewer will reasonably ask whether careful prompt engineering would do the same work as echoing, making the ordering story incidental. We test this with GEPA (Agrawal et al., 2025), a reflective prompt optimizer, and use it as a prediction test rather than as a baseline to beat.

The mechanism predicts an asymmetry. GEPA may rewrite the system message text but cannot move the image, so it cannot repair a read-out failure caused by token order. It should find recoverable headroom under *STI*, where read-out is broken, and little or nothing under *SIT*, where it is not. That is what happens (Table 5, Appendix H). Under *STI*, GEPA improves all three benchmarks:

Table 5: GEPA prompt optimization under two orderings, Qwen3-VL-8B, on held-out evaluation sets. GEPA finds real gains under STI and nothing under SIT, as the read-out diagnosis predicts. It never reaches the SIT baseline it is competing with.

Benchmark	Order	N	Accuracy		Δ	Gap closed
			Seed	GEPA		
POPE	STI	8,800	86.41	87.25	+0.84	45%
	SIT	8,800	88.27	88.27	+0.00	n/a
VQAv2	STI	10,000	86.11	88.13	+2.02	83%
	SIT	10,000	88.55	88.55	+0.00	n/a
NaturalBench	STI	7,000	74.74	75.79	+1.04	26%
	SIT	7,000	78.80	78.39	-0.41	n/a

POPE 86.41 $\rightarrow$ 87.25, VQAv2 86.11 $\rightarrow$ 88.13, NaturalBench 74.74 $\rightarrow$ 75.79. Under SIT, the same optimizer with the same budget returns the seed prompt unchanged on POPE and VQAv2, both exact ties, and regresses NaturalBench by 0.41 by overfitting its validation split.

Two further numbers matter. First, SIT’s baseline is higher than STI’s on every benchmark (88.27 vs 86.41, 88.55 vs 86.11, 78.80 vs 74.74), so the asymmetry is a headroom effect and not a quirk of the optimizer. Second, GEPA’s best STI prompt never reaches the plain SIT baseline: it closes 26% to 83% of the STI-to-SIT gap and stops there. Its optimized prompts add 26, 32 and 71 tokens per query, less than even an eighth-resolution echo, but its search costs 157,725 to 982,343 tokens once plus a separate reflection model. Rewording the prompt recovers part of what bad ordering costs. Moving or echoing the image recovers the rest.

Finding. Prompt optimization behaves exactly as the read-out diagnosis predicts: GEPA gains under STI (POPE +0.84, VQAv2 +2.02, NaturalBench +1.04) and finds nothing under SIT (two exact ties and -0.41), because SIT has less headroom to recover. It closes 26% to 83% of the STI-to-SIT gap and never reaches the SIT baseline, so rewording is not a substitute for fixing token order.

6 DISCUSSION

Three results here constrain what the echo can be. It survives being reduced to $\frac{1}{64}$ of its pixels, it survives being turned upside down, and it fails precisely where the visual context is already long. None of that is consistent with the echo supplying a second, better look at the image. All of it is consistent with the echo supplying a read-out surface: tokens the answer position can reach, whose content barely matters and whose position matters entirely. The one place content does matter, RF20, is the one place the evidence is a small region rather than a whole scene, and there degrading the echo costs 1.76 F1.

Implications beyond prompt ordering. For benchmarking, an unreported prompt-layout choice is a confound of the same size as the model differences benchmarks are built to detect: question-first versus question-last moves CV-Bench by 18.01 points and one RF20 category by 47.35, so two implementations of “the same” prompt can differ by more than the gap being measured. For deployed and agentic systems, pipelines that prepend a goal to a growing context and then append retrieved images produce question-first layouts by construction and pay for it silently; the fix costs 13 tokens on video and about 107 on stills. For training recipes, the failure lives in the read-out path rather than in perception, which suggests instruction tuning that varies section order rather than fixing one canonical template, and the Gemma models, which show no paradox, suggest it is trainable away.

Limitations. The diagnosis this paper builds on is established on Qwen3-VL-8B and does not hold for either Gemma model, so the read-out account describes a common failure mode rather than a property of the architecture; image echoing still helps the Gemma models (Appendix C), but the mechanism we recap cannot be the reason there. The repair is also not uniformly right. It does not beat question-last on POPE (88.12 against 88.35 F1), and on video it loses on three of four datasets,

which is why we recommend question echoing rather than image echoing there. Question echoing alone is not reliably better than simply asking the question last: on NaturalBench `STIT` scores 35.00 against `SIT`'s 35.05 ($p = 1.00$), so its role is to repair question-first prompting, not to improve on question-last prompting. Finally, the knockout result depends on restricting the query side to the answer position; widening it does not reproduce the question-side half of the dissociation, and the span-size control is itself large there, so we treat the broader scope as uninformative rather than as contrary evidence (Appendix J).

ETHICS STATEMENT

This work studies prompt layout for existing, publicly released vision-language models and introduces no new model, no new data collection and no human subjects. All benchmarks used are public research datasets used for their stated purpose. The intervention we study makes models more accurate without fine-tuning, which lowers the compute cost of deployment but does nothing to correct the biases those models already carry; a more accurate answer is not a fairer one. RF20 includes surveillance-adjacent detection domains, and improving presence detection there carries the usual dual-use risk of object detection systems. We report the settings where our intervention fails, including video and POPE, so that it is not applied where it does not help.

REPRODUCIBILITY STATEMENT

Every accuracy in this paper is read from a per-run result file recording the model, ordering, system prompt, decoding budget and sample count. Section 2 gives the models, benchmarks and protocol; Appendix J records the two configuration details that change the result if missed, namely the knockout query scope and the exclusion of a degenerate Gemma RF20 run. Code, the per-run result files and the figure scripts, which regenerate every figure directly from those files, are provided in the supplementary material, along with the prior workshop paper this work extends. Significance tests use paired McNemar plus a paired bootstrap over 5,000 resamples with a fixed seed.

AI USE DISCLOSURE

We used generative AI tools to help write code and to proofread and edit the paper text. All AI-assisted code and text were reviewed and checked by the authors, who take full responsibility for the content of this work.

REFERENCES

- Manoj Acharya, Kushal Kafle, and Christopher Kanan. TallyQA: Answering complex counting questions. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 2019.
- Lakshya A Agrawal, Shangyin Tan, Dilara Soylu, Noah Ziemis, Rishi Khare, Krista Opsahl-Ong, Arnav Singhvi, Herumb Shandilya, Michael J Ryan, Meng Jiang, Christopher Potts, Koushik Sen, Ion Stoica, Dan Klein, Matei Zaharia, and Omar Khattab. GEPA: Reflective prompt evolution can outperform reinforcement learning. *arXiv preprint arXiv:2507.19457*, 2025.
- Anonymous. Title withheld to preserve anonymity, 2026. Prior workshop paper by the present authors. The full paper is included in the supplementary material for reviewers.
- Nora Belrose, Zach Furman, Logan Smith, Danny Halawi, Igor Ostrovsky, Lev McKinney, Stella Biderman, and Jacob Steinhardt. Eliciting latent predictions from transformers with the tuned lens. *arXiv preprint arXiv:2303.08112*, 2023.
- Lin Chen, Jinsong Li, Xiaoyi Dong, Pan Zhang, Yuhang Zang, Zehui Chen, Haodong Duan, Jiaqi Wang, Yu Qiao, Dahua Lin, and Feng Zhao. Are we on the right way for evaluating large vision-language models? *arXiv preprint arXiv:2403.20330*, 2024.
- Xingyu Fu, Yushi Hu, Bangzheng Li, Yu Feng, Haoyu Wang, Xudong Lin, Dan Roth, Noah A Smith, Wei-Chiu Ma, and Ranjay Krishna. BLINK: Multimodal large language models can see but not perceive. In *European Conference on Computer Vision (ECCV)*, 2024.

- Mor Geva, Jasmijn Bastings, Katja Filippova, and Amir Globerson. Dissecting recall of factual associations in auto-regressive language models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2023.
- Yash Goyal, Tejas Khot, Douglas Summers-Stay, Dhruv Batra, and Devi Parikh. Making the V in VQA matter: Elevating the role of image understanding in visual question answering. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017.
- Yunseok Jang, Yale Song, Youngjae Yu, Youngjin Kim, and Gunhee Kim. TGIF-QA: Toward spatio-temporal reasoning in visual question answering. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017.
- Baiqi Li, Zhiqiu Lin, Wenxuan Peng, Jean de Dieu Nyandwi, Daniel Jiang, Zixian Ma, Simran Khanuja, Ranjay Krishna, Graham Neubig, and Deva Ramanan. NaturalBench: Evaluating vision-language models on natural adversarial samples. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024a.
- Kunchang Li, Yali Wang, Yinan He, Yizhuo Li, Yi Wang, Yi Liu, Zun Wang, Jilan Xu, Guo Chen, Ping Luo, Limin Wang, and Yu Qiao. MVBench: A comprehensive multi-modal video understanding benchmark. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024b.
- Yifan Li, Yifan Du, Kun Zhou, Jinpeng Wang, Wayne Xin Zhao, and Ji-Rong Wen. Evaluating object hallucination in large vision-language models. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2023.
- Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. Locating and editing factual associations in GPT. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.
- nostalgebraist. Interpreting GPT: The logit lens. LessWrong, 2020.
<https://www.lesswrong.com/posts/AcKRB8wDpdaN6v6ru/interpreting-gpt-the-logit-lens>.
- Tristan Thrush, Ryan Jiang, Max Bartolo, Amanpreet Singh, Adina Williams, Douwe Kiela, and Candace Ross. Winoground: Probing vision and language models for visio-linguistic compositionality. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2022.
- Shengbang Tong, Ellis Brown, Penghao Wu, Sanghyun Woo, Manoj Middepogu, Sai Charitha Akula, Jihan Yang, Shusheng Yang, Adithya Iyer, Xichen Pan, Ziteng Wang, Rob Fergus, Yann LeCun, and Saining Xie. Cambrian-1: A fully open, vision-centric exploration of multimodal LLMs. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.
- Junbin Xiao, Xindi Shang, Angela Yao, and Tat-Seng Chua. NExT-QA: Next phase of question-answering to explaining temporal actions. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021.
- Dejing Xu, Zhou Zhao, Jun Xiao, Fei Wu, Hanwang Zhang, Xiangnan He, and Yueting Zhuang. Video question answering via gradually refined attention over appearance and motion. In *Proceedings of the 25th ACM International Conference on Multimedia*, 2017.

A RF20 PRESENCE TASK: THE YES/NO VARIANT

Besides the detection task of Section 4.2, RF20’s images support a yes/no presence question (“is a <class> in the image?”), 24,449 questions over 20 datasets in 7 categories. This variant has the same answer format as NaturalBench, POPE and Winoground, and it behaves like them: question-first hurts on 7 of 7 categories, by 11.36 points on average. On Lab Imaging it takes accuracy from 86.29 to 38.94, a 47.35 point loss, and both echo orderings recover the category past the question-last baseline. The contrast with Table 3, where the same images under a detection objective show the opposite sign, is the cleanest evidence we have that the paradox tracks the answer format rather than the imagery.

Table 6: RF20 presence task, accuracy by category, Qwen3-VL-8B, 24,449 questions. Question-first hurts on 7 of 7 categories, opposite in sign to the detection results on the same images.

Category	N	SIT	STI	STIT	SITIT	Δ STI	Δ SITIT
Document	10,200	80.94	76.20	80.30	84.19	-4.74	+3.25
Misc	4,114	85.71	82.50	85.90	85.05	-3.21	-0.66
Industrial	2,958	82.45	68.19	81.54	82.18	-14.26	-0.27
Lab Imaging	2,822	86.29	38.94	91.71	90.93	-47.35	+4.64
Sport	2,654	49.06	48.61	42.99	51.96	-0.45	+2.90
Flora/Fauna	1,592	86.56	82.54	87.44	86.75	-4.02	+0.19
Aerial	109	94.50	88.99	89.91	94.50	-5.51	+0.00
<i>Mean over 7</i>						-11.36	+1.44

B ATTENTION KNOCKOUT IN FULL

Qwen3-VL-8B, 250 NaturalBench pairs, query side restricted to the answer position. `ko_question` severs the attention edges from the answer position to the question span, `ko_image` to the image span, and `ko_random` to an equal-length random span of pre-image text as a span-size control.

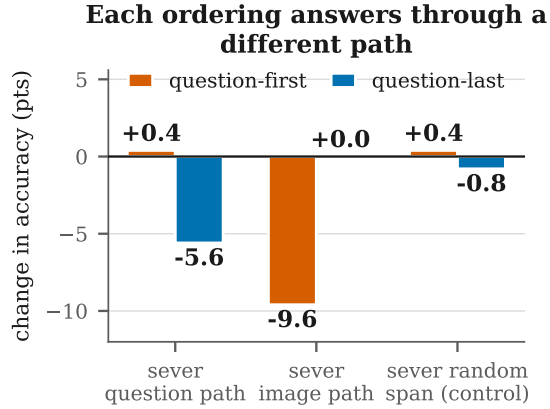


Figure 5: Each ordering answers through a different path. Bars are the change in accuracy against the unmodified run (57.6 for question-first, 58.8 for question-last). Severing answer-to-question hurts only question-last; severing answer-to-image hurts only question-first; the random span-size control moves both by at most 0.8 points.

The paired bootstrap on the question-last question knockout gives $\Delta = 0.056$ with a 95% CI of $[+0.0199, +0.096]$, so the interval excludes zero.

C CROSS-MODEL LADDERS IN FULL

D SIGNIFICANCE TESTS IN FULL

Paired on group index (NaturalBench) or example index (Winoground), exact two-sided McNemar plus a paired bootstrap 95% CI over 5,000 resamples, seed 0. Accuracies are group accuracy.

E READ-OUT PROBE AND DECISION LAYER IN FULL

Qwen3-VL-8B, 36 layers. The disagreement set is the 150 NaturalBench pairs where STI is wrong and SIT is right; the decision-layer analysis uses 300 pairs. a_q and a_{img} are attention mass from the answer position onto the question and image spans, averaged over heads.

Table 7: NaturalBench ladder across five models (1,900 groups). Four of five models lose accuracy when the question moves before the image. Missing entries were not run. LLaVA-1.5’s question-first group accuracy is exactly zero.

Model	Metric	SIT	STI	STIT	SITIT	Δ (STI-SIT)
Qwen3-VL-8B	pair acc.	79.21	75.29	78.95	80.36	-3.92
	group acc.	35.05	26.95	35.00	37.37	-8.10
Qwen2.5-VL-7B	pair acc.	76.05	64.95	71.83	75.71	-11.10
	group acc.	27.74	10.16	19.11	27.63	-17.58
InternVL3-8B	pair acc.	79.63	75.61	79.76	80.80	-4.02
	group acc.	35.53	27.63	36.47	39.32	-7.90
LLaVA-1.5	pair acc.	66.87	50.26	66.29	n/a	-16.61
	group acc.	12.32	0.00	12.05	n/a	-12.32
Gemma-3-27B	pair acc.	73.34	73.21	74.64	74.61	-0.13
	group acc.	22.63	23.21	25.26	25.53	+0.58

Table 8: Cross-model check on the four benchmarks run for all three models. The question-first paradox does not replicate on either Gemma model, yet image echoing is still the best ordering on 7 of 8 model-benchmark pairs.

Model	Benchmark	N	Ordering				Δ vs SIT	
			SIT	STI	STIT	SITIT	STI	SITIT
Gemma-3-27B	BLINK	793	44.51	43.51	42.75	44.89	-1.01	+0.38
	MMVP	300	72.67	72.33	72.00	74.67	-0.33	+2.00
	TallyQA	38,589	63.00	67.60	68.93	69.97	+4.60	+6.96
	VQAv2	214,354	64.77	62.07	63.04	62.07	-2.70	-2.70
	Mean						+0.14	+1.66
Gemma-4-31B	BLINK	793	54.48	56.12	56.12	56.49	+1.64	+2.02
	MMVP	300	84.67	86.00	86.67	87.33	+1.33	+2.67
	TallyQA	2,000	80.65	80.50	81.20	82.60	-0.15	+1.95
	VQAv2	3,000	73.50	74.66	75.65	76.00	+1.16	+2.50
	Mean						+1.00	+2.28

$P(\text{correct})$ over the {Yes, No} decision on the 300-pair disagreement set. All four orderings sit at ≈ 0.50 through layer 25, so only layers 26 to 35 are shown.

Layer	26	27	28	29	30	31	32	33	34	35
STI	0.480	0.456	0.440	0.449	0.495	0.429	0.398	0.222	0.099	0.152
SIT	0.593	0.665	0.680	0.663	0.529	0.691	0.728	0.824	0.875	0.851
SITIT	0.624	0.691	0.710	0.704	0.546	0.704	0.722	0.827	0.819	0.827
STIT	0.526	0.513	0.523	0.521	0.500	0.532	0.533	0.547	0.553	0.570

F PER-PATCH COSINE TO AN IMAGE-ONLY FORWARD PASS

Layer 18, images resized to 448×448 , $14 \times 14 = 196$ patches, Qwen3-VL-8B. Mean per-patch cosine between each ordering’s image-token hidden states and a forward pass on the image alone.

Sanity checks: $\cos(\text{base}, \text{IST}) = 0.999997$ and $\cos(\text{STI}, \text{STIT}) = 0.999992$ at $n = 7,600$.

Table 9: Significance for consecutive rungs of the ladder and for the headline comparisons. STI versus SIT is highly significant for Qwen3-VL-8B on both benchmarks and absent for Gemma-3-27B. SITIT versus SIT is significant on Winoground for both models.

Benchmark	Model	Comparison	acc A	acc B	Δ (95% CI)	McNemar p
NaturalBench	Qwen3-VL-8B	STI vs SIT	0.269	0.351	+0.081 [+0.058, +0.103]	1.3×10^{-12}
		SIT vs SITIT	0.351	0.350	-0.001 [-0.022, +0.020]	1.00
		STIT vs SITIT	0.350	0.374	+0.024 [+0.005, +0.043]	1.6×10^{-2}
		SITIT vs SITIT_rev	0.374	0.374	+0.000 [-0.010, +0.011]	1.00
	Gemma-3-27B	STI vs SIT	0.232	0.226	-0.006 [-0.026, +0.015]	0.62
		SIT vs SITIT	0.226	0.253	+0.026 [+0.006, +0.046]	1.4×10^{-2}
		STIT vs SITIT	0.253	0.255	+0.003 [-0.015, +0.021]	0.82
	Winoground	STI vs SIT	0.223	0.318	+0.095 [+0.050, +0.140]	7.7×10^{-5}
		SIT vs SITIT	0.318	0.375	+0.057 [+0.015, +0.102]	1.2×10^{-2}
		SIT vs SITIT	0.318	0.403	+0.085 [+0.045, +0.125]	2.4×10^{-5}
		SIT vs SITIT_rev	0.223	0.410	+0.187 [+0.140, +0.235]	1.2×10^{-13}
	Gemma-3-27B	STI vs SIT	0.212	0.207	-0.005 [-0.052, +0.043]	0.92
		SIT vs SITIT	0.207	0.253	+0.045 [+0.000, +0.087]	6.3×10^{-2}
		SIT vs SITIT	0.207	0.318	+0.110 [+0.070, +0.152]	8.1×10^{-7}

Table 10: Read-out probe on the disagreement set. STI peaks lower on the question and much higher on the image, and the mid-to-late band ratio inverts.

Ordering	probe acc	max a_q (layer)	max a_{img} (layer)	$a_q:a_{img}$ @L24	final $P(gt)$
STI	0.073	0.0676 (23)	0.2367 (21)	0.424	0.1346
SIT	0.887	0.1479 (23)	0.0942 (18)	1.065	0.8485
STIT	0.600	0.1437 (23)	0.1448 (21)	0.986	0.5912
SITIT	0.847	0.1286 (23)	0.0647 (26)	1.544	0.8374
SITIT_rev	0.867	0.1218 (23)	0.0617 (26)	1.624	0.8503

G EXACT TOKEN COST PER ORDERING

H GEPA CONFIGURATION AND TRAINING COST

GEPA (Agrawal et al., 2025) optimizes the system message only. The task model is Qwen3-VL-8B; the reflection model is Gemma-3-27B on a separate GPU, since reusing the task model as its own reflector produced proposals that never passed GEPA’s strict-improvement acceptance test. Training and validation examples are disjoint from the held-out evaluation set.

I LOGIT-LENS WORD DIVERGENCE BETWEEN STI AND IST

To ask whether question-first and image-first prompting diverge in the words they are internally predicting, we decode the top-1 token at every (layer, generation step) position under both orderings on 100 samples, discard positions whose two words are identical or WordNet synonyms, and discard any position where either side is not a dictionary word, since intermediate layers frequently project onto subword fragments.

Genuine word changes are rare and concentrated: 8 across 100 samples, of which 6 occur at layer 16, one at layer 8 and one at layer 20, with none at layers 0, 4, 12, 24 or 27. Divergence at the level of decoded output words is therefore sparse and mid-network, consistent with the read-out account in Section 3.3, where the two orderings are indistinguishable at the decision level until late and differ mainly in routing rather than in vocabulary.

Table 11: The question rewrites the image tokens. IST is the image-only baseline by construction; STIT is identical to STI because causal masking prevents a later question copy from affecting earlier patches.

Ordering	n	mean cos	median	min
IST	7,600	0.999998	1.000000	0.999919
SIT	30	0.911017	0.921399	0.722478
STI	7,600	0.862723	0.873233	0.614870
STIT	7,600	0.862727	0.873237	0.614835

Table 12: Exact token cost per query, mean over 20 images spanning RF20’s constituent datasets, Qwen3-VL-8B. Δ is against the STI baseline. STI and SIT contain the same tokens in a different order, so their totals are identical. Repeating the question is nearly free; repeating the image is not, but its cost falls with echo area because vision tokens scale with pixels.

Prompt	Total	Image	Text	Δ total	Δ image	Δ text
STI (baseline)	3385.8	3335.8	50.0	n/a	n/a	n/a
STIT (question echo)	3398.8	3335.8	63.0	+13.0	+0.0	+13.0
SITIT, echo 1	6737.5	6671.5	66.0	+3351.8	+3335.8	+16.0
SITIT, echo $\frac{1}{2}$	4260.7	4194.7	66.0	+874.9	+858.9	+16.0
SITIT, echo $\frac{1}{4}$	3637.6	3571.6	66.0	+251.8	+235.8	+16.0
SITIT, echo $\frac{1}{8}$	3492.6	3426.6	66.0	+106.8	+90.8	+16.0

J REPRODUCTION NOTES

Every accuracy in this paper is read from a per-run result file that records the model, the ordering, the system prompt, the decoding budget and the sample count. Two details are worth flagging for anyone re-running the analysis.

The attention knockout must restrict the query side to the answer position. With the query side widened to every position after the question, the question-side half of the dissociation does not reproduce (severing answer-to-question raises SIT by 2.8 points, CI $[-0.084, +0.028]$), and the random-span control is itself worth 10 points on STI, so that configuration is uninformative rather than contradictory.

The Gemma-3-27B RF20 run is degenerate and we exclude it from all RF20 claims. It answers “yes” to roughly 96% of items under every ordering (`yes_ratio` 0.955 to 0.964, recall 0.979 to 0.991), so its accuracy sits at the positive base rate and the spread across orderings is 0.003.

Table 13: GEPA search budget and cost, *STI* ordering. Training tokens are a one-time cost; the per-query cost is the prompt-length delta in Table 14.

Benchmark	train	val	metric calls	training tokens	wall clock (s)
POPE	100	60	400	157,725	124.3
VQAv2	150	80	1,008	396,253	232.4
NaturalBench	250	120	1,540	982,343	528.7

Table 14: Prompt-length cost of GEPA’s optimized system messages, in tokens per query. Under *SIT* on POPE and VQAv2 the optimizer returned the seed prompt, so the delta is zero.

Benchmark	Order	seed	optimized	Δ
POPE	<i>STI</i>	315	341	+26
	<i>SIT</i>	316	316	+0
VQAv2	<i>STI</i>	313	345	+32
	<i>SIT</i>	314	314	+0
NaturalBench	<i>STI</i>	216	287	+71
	<i>SIT</i>	217	381	+164